① compute orthogonal projection of a vector (1, 7) onto the line through the vector (-4, 2) and the origin.

⇒ let w = line through $(-4, 2)$ & $(0, 0)$

$= \text{span}\{\underbrace{(-4, 2)}_{w}\}$

let $v = (1, 7)$

$$\text{proj}_w v = \frac{v \cdot w}{w \cdot w} w$$

$v \cdot w = (1, 7) \cdot (-4, 2)$

$= -4 + 14 = \underline{\underline{10}}$

$\left(\cancel{w \cdot w = (1, 7) \cdot (1, 7)} \right.$
$\left. \cancel{= 1 + 49 = \underline{\underline{50}}} \right)$

$w \cdot w = +16 + 4$

$= \underline{\underline{20}}$

$$= \frac{10}{20}(-4, 2)$$

$$= \frac{1}{2}(-4, 2)$$

$$= \underline{\underline{(-2, 1)}}$$

② Let $y = (7,6)$ and $u = (4,2)$ find the orthogonal projection of y on u and also write y as sum of two vectors: one is span$\{u\}$ & one is orthogonal to u.

$\Rightarrow$ $y = (7,6)$ $\quad u = (4,2)$

$$\text{proj}_u y = \frac{y \cdot u}{u \cdot u} u$$

$y \cdot u = 28 + 12 = 40$

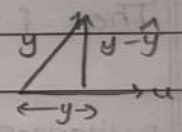
$u \cdot u = 16 + 4 = 20$

$$= \frac{40}{20}(4,2)$$

$$= 2(4,2)$$

$\hat{y} = (8,4) \in \text{span}\{u\}$

$y = \hat{y} + (y - \hat{y}) \longrightarrow$



$y - \hat{y} = (7,6) - (8,4)$

$= (-1,2) =$ orthogonal to u

$y = \hat{y} + (y - \hat{y})$

$= (8,4) + (-1,2)$

$y = \underline{\underline{(7,6)}}$

H.W ③ Compute orthogonal projection of $y = (2,3)$ on the line through the point $u = (4,-7)$ then write y as the sum of two orthogonal vectors: one as span$\{u\}$ and one orthogonal to u.

Grahm - schmidt process:

The gram schmidt process is a simple algorithm for pocketing an orthogonal / orthonormal basis for any non-zero subspace of R.

Given a basis $\{x_1, x_2, \ldots, x_p\}$ for a non-zero subspace W of R^n define,

$$v_1 = x_1$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2$$

$$\vdots$$

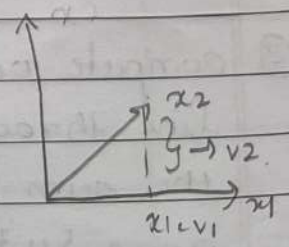
$$v_p = x_p - \frac{x_p \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_p \cdot v_2}{v_2 \cdot v_2} - \ldots - \frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}}$$

Then $\{v_1, v_2, \ldots, v_p\}$ is orthogonal vector

Orthonormal Basis:

Orthonormal basis is constructed easily from an orthogonal basis $\{v_1, v_2, \ldots, v_p\}$ simply normalize all the v_k.

Q] Let $W = \text{span}\{x_1, x_2\}$ where $x_1 = (3, 6, 0)$, $x_2 = (1, 2, 2)$. Find an orthogonal basis of W by using Grahm Schmidt process.



$\Rightarrow$ Let $v_1 = x_1$

$$v_2 = x_2 - \text{proj}_{v_1} x_2$$

$$= x_2 - \frac{v_1 \cdot x_2}{v_1 \cdot v_1} v_1$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{\cancel{15}\,1}{\cancel{45}\,3} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

$x_2 \cdot v_1 = x_2 \cdot x_1$
$= 3 + 12$
$= 15$

$$= \begin{bmatrix}1\\2\\2\end{bmatrix} - \begin{bmatrix}1\\2\\0\end{bmatrix} = \begin{bmatrix}0\\0\\2\end{bmatrix}$$

$\therefore \{v_1, v_2\}$ is an orthogonal basis of W

2] Let $W = \text{span}\{x_1, x_2, x_3\}$ where ~~$x_1 = (1,1,0)$~~ $x_1 = (1,1,1,1)$, $x_2 = (0,1,1,1)$ & $x_3 = (0,0,1,1)$. find orthonormal basis of W by using Gram schmidt process.

$\Rightarrow$ Let $x_1 = v_1$

$$v_2 = x_2 - \text{proj}_{v_1} x_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} \cdot v_1$$

$0+1+1+1$
$= 3$

$$= \begin{bmatrix}0\\1\\1\\1\end{bmatrix} - \frac{3}{4} \begin{bmatrix}1\\1\\1\\1\end{bmatrix}$$

$$= \begin{bmatrix}-3/4\\1/4\\1/4\\1/4\end{bmatrix} = 1/4 \begin{bmatrix}-3\\1\\1\\1\end{bmatrix}$$

$$v_3 = x_3 - \text{proj}_{v_1} x_3 - \text{proj}_{v_2} x_3$$

$$= \begin{bmatrix}0\\0\\1\\1\end{bmatrix} - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2$$

$$v_3 = \begin{bmatrix}0\\0\\1\\1\end{bmatrix} - \frac{2}{4}\begin{bmatrix}1\\1\\1\\1\end{bmatrix} - \cancel{\frac{2}{3}\begin{bmatrix}0\\1\\1\\1\end{bmatrix}} - \frac{1/4\,(2)}{1/4 \times 1/4\,(12)} \cancel{1/4} \begin{bmatrix}-3\\1\\1\\1\end{bmatrix}$$

$$\cancel{\frac{1/4 \times 1/4 \times 2 \times 1}{1/4 \times 1/4 \times 1/4}}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{6}\begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 - \frac{1}{2} + \frac{1}{2} \\ 0 - \frac{1}{2} - \frac{1}{6} \\ 1 - \frac{1}{2} - \frac{1}{6} \\ 1 - \frac{1}{2} - \frac{1}{6} \end{bmatrix} = \begin{bmatrix} 0 \\ -4/6 \\ 1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore \{v_1, v_2, v_3\}$ is an orthogonal basis of W.

Let $u_1 = \dfrac{v_1}{\|v_1\|} = \dfrac{(1,1,1,1)}{\sqrt{1^2+1^2+1^2+1^2}} = \dfrac{1}{\sqrt{4}}(1,1,1,1)$

$$= \frac{1}{2}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$u_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{(1/4)^2\left[(-3)^2+1^2+1^2+1^2\right]}} \times \frac{1}{4}\begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\frac{1}{4}(\sqrt{12})} \times \frac{1}{4}\begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\frac{1}{4} \times 2\sqrt{3}} \times \frac{1}{4}\begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2\sqrt{3}}\begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$u_3 = \frac{v_3}{\|v_3\|}$$

$$= \frac{1}{\sqrt{\left(\frac{1}{3}\right)^2 \sqrt{[0+4+1+1]}}} \times \frac{1}{3}\begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$$\frac{1}{\cancel{\frac{1}{3}}(\sqrt{6})} \times \cancel{\frac{1}{3}}\begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$$u_3 = \frac{1}{\sqrt{6}}\begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore \{u_1, u_2, u_3\}$ is an orthonormal basis

3] Let $W = \text{span}\{x_1, x_2, x_3\}$ where $x_1 = (1,1,1,1)$, $x_2 = (1,-1,2,3)$, $x_3 = (1,2,-3,-4)$. construct an orthogonal basis of W.

$\Rightarrow$ $v_1 = x_1$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} \cdot v_1$$

$x_2 \cdot x_1 = \cancel{1} + (-\cancel{1}) + 2 + 3$
$= 5$.

$x_1 \cdot x_1 = 4$

$$= \begin{bmatrix} 1 \\ -1 \\ 2 \\ 3 \end{bmatrix} - \frac{5}{4}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} -1/4 \\ -9/4 \\ 3/4 \\ 7/4 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -1 \\ -9 \\ 3 \\ 7 \end{bmatrix}$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2$$

$$= x_3 - \frac{(-4)}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{1/4(-56)}{1/16(1+81+9+49)} \frac{1}{4} \begin{bmatrix} -1 \\ -9 \\ 3 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ -3 \\ -4 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{2}{5} \begin{bmatrix} -1 \\ -9 \\ 3 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} 1 + 1 - 2/5 \\ 2 + 1 - 18/5 \\ -3 + 1 + 6/5 \\ -4 + 1 + 14/5 \end{bmatrix} = \begin{bmatrix} 8/5 \\ -3/5 \\ -1/5 \\ -1/5 \end{bmatrix} =$$

$\{v_1, v_2, v_3\}$ is an orthogonal basis of W.

4] If $W = \text{span}\{x_1, x_2\}$, where $x_1 = (2, -5, 1)$, $x_2 = (4, -1, 2)$. construct orthonormal basis for W.